
Understanding max_tokens in the Anthropic API

1. What is max_tokens?

max_tokens sets the upper limit on how many tokens the model can generate in its response. It is a hard cap - the model stops generating the moment it hits that count, even mid-sentence.

2. Why Anthropic Requires It Explicitly

Unlike OpenAI, where max_tokens is optional and defaults to a model-defined limit, Anthropic requires max_tokens to be explicitly set in every API call. The request will be rejected with a validation error if it is omitted.

3. What is a Token?

Roughly 1 token $\approx$ 3/4 of an English word. As a rule of thumb:

- 1,024 tokens $\approx$ 750 words
- 4,096 tokens $\approx$ 3,000 words

Tokens include both the characters of the word and surrounding whitespace or punctuation, so the exact ratio varies by language and content.

4. What Happens When the Limit is Reached?

If the response is cut short by the token limit, the stop_reason field in the response object will be "max_tokens" instead of the normal "end_turn". You should always check this field to detect truncated responses:

```
if response.stop_reason == "max_tokens":  
    print("Warning: response was truncated!")
```

5. Recommended Values by Use Case

Use Case	Suggested max_tokens	Approx. Word Count
Short factual Q&A	256 - 512	approx. 190 - 380 words
Summaries / explanations	1024 - 2048	approx. 750 - 1,500 words
Long-form writing / code	4096 - 8192	approx. 3,000 - 6,000 words
Maximum (claude-sonnet-4-6)	8192	approx. 6,000 words

6. Practical Note for This Script

In llm_call_with_api.py, the question asked is "What is the capital of France?" - a one-word answer. The max_tokens value of 1,024 is far more than needed. A value of 256 would be perfectly sufficient and is better practice for cost and latency efficiency.